

Education

- Mila and Polytechnique Montréal**, Montréal, Canada *Aug'22-Present*
Expected: Jan 2027
PhD in Computer Engineering; Advisor: [Sarath Chandar](#)
- Indian Institute of Technology Madras**, Chennai, India *Jul'17-Jul'22*
CGPA: 9.31/10
BS in Biological Sciences and MTech in Data Science; Minor: Computational Biology

Research Areas and Interests

Reinforcement Learning, Large Language Models, AI-driven Material and Drug Discovery, Agentic Systems, Geometric Deep Learning, and Computational Biology

Coursework & Skills

Courses: Geometry and Generative Models, Reinforcement Learning, Representation Learning, Robot Learning, Parameter and State Estimation, Parallel Scientific Computing, Algorithmic Approaches to Computational Biology, Pattern Recognition and Machine Learning
Skills: Python (PyTorch, Tensorflow, JAX), R, MATLAB, C/C++ (OpenMP, MPI, OpenACC), Matter Modeling (DFT)

Professional Experience

- Software Engineering PhD Intern, Google** *May'26-Aug'26*
Location: Bengaluru, India
– Designing agentic systems for automated semantic discovery in Google Cloud's global infrastructure team
– **Skills:** Agentic system design frameworks (e.g., Google ADK), testing and evaluating agents, NLP (semantic search, vector databases), software engineering

Research Experience

- Reinforcement learning for LLMs** *Sep'25-Present*
Mentors: [Sarath Chandar](#), [Saurav Jha](#)
– Improving exploration with intrinsic rewards in RL-based fine-tuning of LLMs
- AI-automated CAD object generation** *Jun'24-Present*
Mentors: [Sarath Chandar](#), [Jay Pathak](#), [Quentin Fournier](#)
– Collaboration with Ansys: LLMs for automated generation of 3D CAD objects
– Published CADmium, an open-source dataset and LLM fine-tuning approach at TMLR [[Paper](#), [Code](#), [Website](#)]
– Currently working on LLM fine-tuning for CAD editability
- Reinforcement learning for material design** *Aug'22-Present*
Mentors: [Sarath Chandar](#), [Mathieu Reymond](#), [Santiago Miret](#), [Mariano Phielipp](#)
– Project with Intel on offline and online RL approaches for generating new crystal structures
– Integrated first-principles DFT with conservative Q-learning – accepted at [MoML 2023](#), [AI4Mat](#) at [NeurIPS 2023](#), and [Digital Discovery](#) [[Paper](#), [Code](#)]
– Released CrystalGym, first online RL environment for material discovery – Spotlight at AI4Mat-ICLR 2025 [[Paper](#), [Code](#)]
– Currently working on symmetry-aware LLM fine-tuning with RL for crystal generation – AI4Science-ICML 2026 [[Paper](#)]
- Master's Thesis: Graph generative models for binding site-specific molecule generation** *Aug'21-Jun'22*
Guides: [Balaraman Ravindran](#), [Karthik Raman](#), IIT Madras [[Thesis](#), [Poster](#)]
– Designed graph VAE models for drug molecule generation; explored RNN/LSTM for node and edge generation with mitigated order dependence
- Analysis of drug response and gene expression data of AML cells** *Jun-Sep'21*
Guide: [Brian Wilhelm](#), Université de Montréal
– Computational methods to identify drug-gene correlations and molecules inducing leukemic cell maturation
- Deep generative models for single-cell gene expression analysis** *May-Jul'20*
Guide: [Hongyu Zhao](#), Yale University (Virtual)
– Evaluated VAEs and other unsupervised deep learning for single-cell gene expression data analysis
- RNA-seq data analysis of human oral squamous cell carcinoma** *May-Jul'19*
Guide: [Debnath Pal](#), Indian Institute of Science Bangalore
– Identified somatic mutations in RNA-seq data of human oral squamous cell carcinoma samples

Projects

- **Effects of visual representation for navigation control tasks**
Robot Learning, Université de Montréal *Jan-Apr'23*
 - Studied effects of contrastive learning and VAE-based pretraining for RL-based visual navigation
- **Incorporating geometry into score-based model for crystal structure design**
Geometry and Generative Models, McGill University *Sep-Dec'22*
 - Incorporated crystal symmetry as inductive bias into generative models for crystal structure design
- **Generating drug-like molecules from gene expression signatures using transformer**
Algorithmic Approaches to Computational Biology, IIT Madras [[Poster](#), [Video](#), [Report](#)] *Sep-Dec'20*
 - Attention-based transformer for *de novo* drug-like molecule generation inducing desired transcriptomic profiles. Poster at MLCSB COSI, [ISMB 2022](#)
- **Parallel analyses of canonic polyadic tensor decomposition algorithm**
Parallel Scientific Computing, IIT Madras [[Report](#), [Code](#)] *Feb-Jun'21*
 - CPU- and GPU-level parallelization of tensor decomposition using OpenMP and OpenACC
- **Deep generative approach to model single-cell data of human embryoid bodies**
Computational Systems Biology, IIT Madras *Jan-Jul'20*
 - Used deep generative VAE model (scVI) to identify biologically relevant cell types of single-cell human embryoid bodies

Awards

- **FRQNT Doctoral Scholarship 2026** – Merit scholarship awarded by Fonds de recherche du Québec
- **PBEEE 2025** – Merit scholarship for international students awarded by Fonds de recherche du Québec
- **Khorana Program for Scholars 2020¹** – Awarded by Department of Biotechnology, Govt. of India (U.S.A research internship fellowship)
- **INSPIRE Scholar** – Awarded by Department of Science and Technology, Government of India

Publications ([Google Scholar](#))

- **Govindarajan, Prashant**, Davide Baldelli, Jay Pathak, Quentin Fournier, and Sarath Chandar. “**CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design.**” arXiv:2507.09792 (2025). *Accepted at TMLR*
- **Govindarajan, Prashant**, Mathieu Reymond, Antoine Clavaud, Mariano Phielipp, Santiago Miret, and Sarath Chandar. “**CrystalGym: A New Benchmark for Materials Discovery Using Reinforcement Learning.**” arXiv:2509.23156 (2025).
- **Govindarajan, Prashant**, Santiago Miret, Jarrod Rector-Brooks, Mariano Phielipp, Janarthanan Rajendran, and Sarath Chandar. “**Learning Conditional Policies for Crystal Design Using Offline Reinforcement Learning.**” Digital Discovery (2024).
- **Govindarajan, Prashant**, Kishalay Das, and Sarath Chandar. “CrysTune: Crystal Generation via Fine-Tuning of Large Language Models on Wyckoff Representations.” ICML 2026 AI for Science Workshop.
- Woo, Kowen, **Prashant Govindarajan**, and Sarath Chandar. “**Benchmarking Machine Learning Potentials for Crystal Structure Relaxation.**” NeurIPS 2025 AI for Science Workshop.
- **Govindarajan, Prashant**, Mathieu Reymond, Santiago Miret, Mariano Phielipp, and Sarath Chandar. “**Crystal Design Amidst Noisy DFT Signals: A Reinforcement Learning Approach.**” AI for Accelerated Materials Design-NeurIPS 2024.
- **Govindarajan, Prashant**, Santiago Miret, Jarrod Rector-Brooks, Mariano Phielipp, Janarthanan Rajendran, and Sarath Chandar. “**Behavioral Cloning for Crystal Design.**” ML4Materials workshop, *ICLR 2023*.

Accepted Posters: AI4Science (ICML 2026 & NeurIPS 2025), [AI4Mat](#) (ICLR 2025, Vienna 2024, NeurIPS 2023 & 2024), ML4Materials (ICLR 2023), [MoML 2023](#), [ISMB 2022](#)

¹Canceled due to Covid-19

Activities

- Teaching** – [Machine Learning](#), Polytechnique Montréal, Fall 2025 | [Reinforcement Learning](#), Polytechnique Montréal, Fall 2023
- [Reinforcement Learning](#), IIT Madras, Spring 2022 | [DSA for Biology](#), IIT Madras, Fall 2021
- Activities** – Information Security Committee at Mila
- Organizer of [AI for Materials](#) reading group at Mila
- Instructor for Chandar Lab’s High School Internship Program
- Volunteer for [Graduate Application Assistance for Underrepresented Students in AI](#)
- Organizer of [MoML 2023](#) and [MoML 2024](#) at Mila
- Social events organizer at [Chandar Lab](#)
- Reviewing** – [TMLR](#) (2025), [AI4Science](#) (NeurIPS 2025, ICML 2026), [AI4Mat](#) workshop (Vienna 2024, NeurIPS 2023 & 2024), [MoML](#) (2023-25), [Deployable AI](#) (AAAI 2023)
- Competitions** – First prize in start-up pitch at [Sciencepreneurship](#), EPFL | Winning team, [MIT COVID-19 Challenge](#) | Finalist, [NYAS Tracking Coronavirus](#)
- Others** – Summer School Acceptances: EEML (2026, online), [Oxford ML](#) (2024, declined), [Sciencepreneurship](#) (2024), [Amii AI Week](#) (2022, online)
- Ultimate Frisbee (NSO, IIT Madras) | Coordinator, Sponsorship team, Shaastra 2019², IIT Madras | Coordinator, Analytics Club, CFI³, IIT Madras

²Technical festival

³Center for Innovation